

Due on tuesday, March 13th, in class

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1. A subatomic particle lives for 6×10^{-7} seconds, i.e., it decays 6×10^{-7} seconds after it is created, as seen from its own reference frame. This particle is created at a speed $v = \frac{4}{5}c$ with respect to earth,

- (a) **[4 pts.]** How long does it live, according to the earth frame? Specify clearly: in which frame is the lifetime the ‘proper’ time?
- (b) **[3 pts.]** How far will the particle travel before it decays, as seen from the earth frame?
- (c) **[3 pts.]** Imagine that the Galilean transformations were true instead of the relativistic (Lorentz) transformations. If this were the case, how far would the particle travel before it decays, as seen from the earth frame?

2. Frame S' moves at speed v_1 with respect to frame S , and frame S'' moves at speed v_2 with respect to frame S' , both in the common x, x', x'' direction. The Lorentz transformations are

$$x' = \gamma_1(x - v_1 t), \quad t' = \gamma_1(t - v_1 x/c^2) \quad \text{with} \quad \gamma_1 = \gamma(v_1) = \frac{1}{\sqrt{1 - v_1^2/c^2}}$$

$$x'' = \gamma_2(x' - v_2 t'), \quad t'' = \gamma_2(t' - v_2 x'/c^2) \quad \text{with} \quad \gamma_2 = \gamma(v_2) = \frac{1}{\sqrt{1 - v_2^2/c^2}}$$

- (a) **[4 pts.]** Express x'' (the coordinate of the S'' frame) in terms of x and t , in the form $x'' = Ax + Bt$. Express A and B in terms of v_1, v_2, γ_1 and γ_2 .
 - (b) **[3 pts.]** If frame S'' moves at speed u with respect to frame S , you would have $x'' = \gamma(u)(x - ut)$. Comparing with $x'' = Ax + Bt$ and using your expressions for A and B , find u . Have you recovered the relativistic velocity addition formula?
 - (c) **[3 pts.]** Verify that your constant A is really equal to $\gamma(u)$, with the u known from the velocity addition formula.
3. **[6 pts]** Show that, if the 4×4 transformation matrix Λ acting on 4-vectors (ct, x, y, z) preserves the Minkowski norm $c^2 t^2 - x^2 - y^2 - z^2$, then it must satisfy $\Lambda^T g \Lambda = g$. Here g is the metric tensor defined in Problem Set 05.

4. Frame Σ' moves at speed v with respect to frame Σ , in the common x, x' direction. The two frames are aligned at time $t = t' = 0$.
- (a) [**5+3 pts.**] A particle has velocity $\vec{u}' = (0, c, 0)$ relative to Σ' . Use the velocity addition formulae to find the velocity of the particle relative to Σ .
Explain how your result is consistent with the constancy of the speed of light.
- (b) [**4 pts.**] A stick is at rest in frame Σ . It lies in the xy plane and makes angle $\pi/4$ with the x -axis. Using the Lorentz transformations or the length contraction formula, find the angle that the stick makes with the x' axis, as observed from the Σ' frame.
5. A stick of proper length L moves past you at speed v . There is a time interval between the front end passing you (first event) and the back end passing you (second event). We are interested in this time interval and in the length of the stick, as seen from various frames.
- (a) [**1 pt.**] Draw snapshots of the two events as seen from your frame, marking with an arrow what is moving (you or the stick).
- (b) [**4 pts.**] Find the length of the stick as seen from your frame, and the time between the two events, as seen from your frame.
- (c) [**1 pts**] Draw snapshots of the two events as seen from the frame of the stick, marking with an arrow what is moving, as seen from this frame.
- (d) [**4 pts.**] Find the length of the stick and the time between the two events, as seen from the frame of the stick.
- (e) [**2 pts.**] In which frame is the time interval the *proper* time interval: your frame or the stick's frame? Explain why.